

OBJECTIVE

To develop an open-source smart emissions monitor for espresso brewing in collaboration with PSU's Coffee and Telesensing Laboratory and Healthy Buildings Research Laboratory. The system combines a modified Gaggia Classic Pro, a custom PCB, environmental sensors, and an emissions enclosure to achieve consistent espresso brewing and repeatable emissions data collection.

OVERVIEW

- Espresso brewing releases volatile organic compounds (VOCs) that affect indoor air quality
- Commercial smart machines are expensive, closed-source, and not designed for research
- Existing open-source modifications lack the sensing and instrumentation researchers need
- The Gaggiuino platform gives us a foundation for embedded control and data logging



Figure 1: Gaggia Espresso Machine



Figure 2: Final design

GAGGIUINO

- STM32 based functional enhancement for Gaggia Espresso Machines
- Temperature control for custom thermal, pressure, and flow profiling
- Bluetooth scale support with automatic brew stop by target weight
- Live brewing data via screen or Wi-Fi interface
- Includes local Wi-Fi connectivity for settings and monitoring

METHODOLOGY

The team started with a Gaggiuino-modified Gaggia Classic Pro espresso machine, adding an emissions enclosure for VOC capture. An ESP32-based "Emissions Control Unit" PCB was designed for sensor reading and fan control within the enclosure.

The board integrates 3 sensors and 3 on-enclosure fans into a single PCB system. Sensor data is packetized and transmitted serially, with future design concepts utilizing network commands to a local webserver. Python tools were developed for data logging.

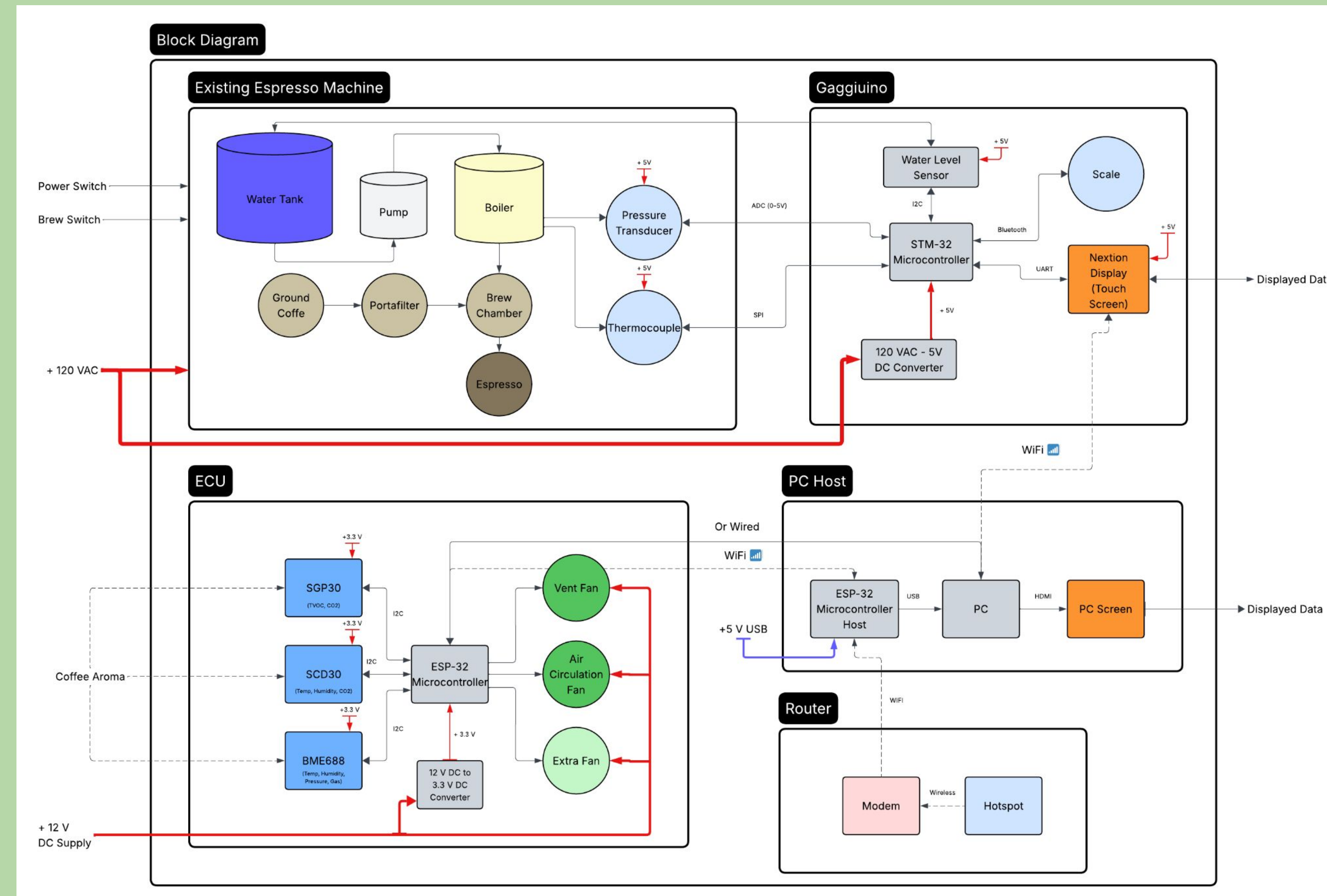


Figure 3: System block diagram

COMPONENTS

Components used in the emissions monitor include sensors, a custom PCB and a ventilation system.

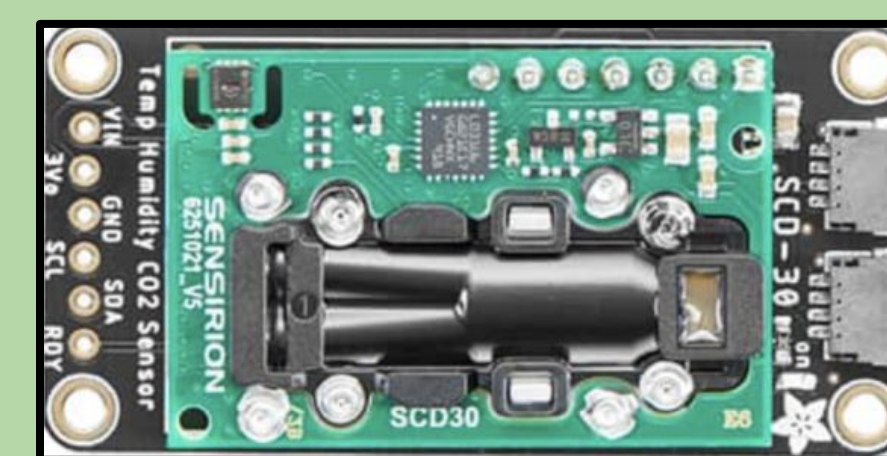


Figure 4: SCD30 sensor

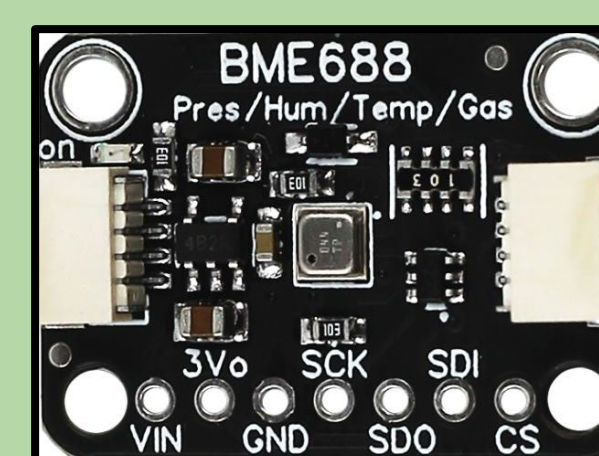


Figure 5: BME688 sensor

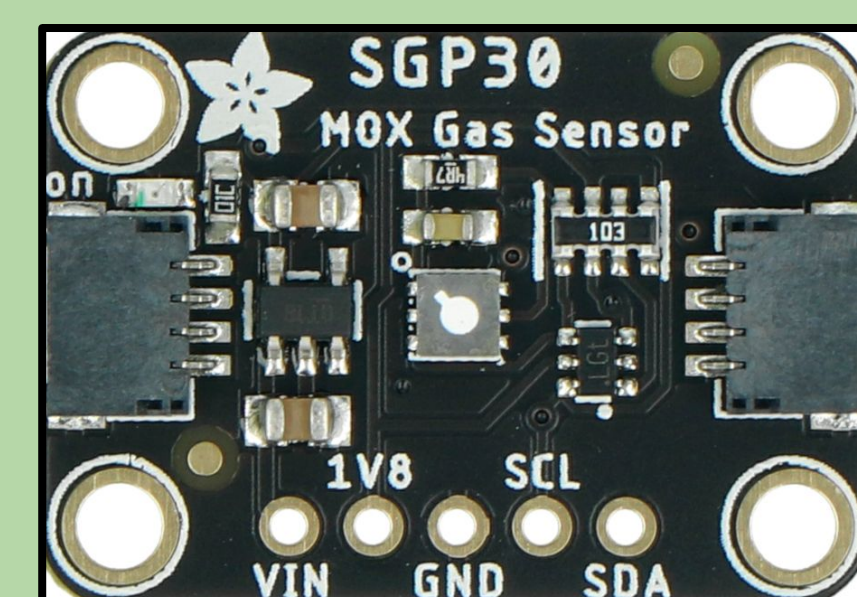


Figure 6: SGP30 sensor



Figure 7: Mixing/External Fan

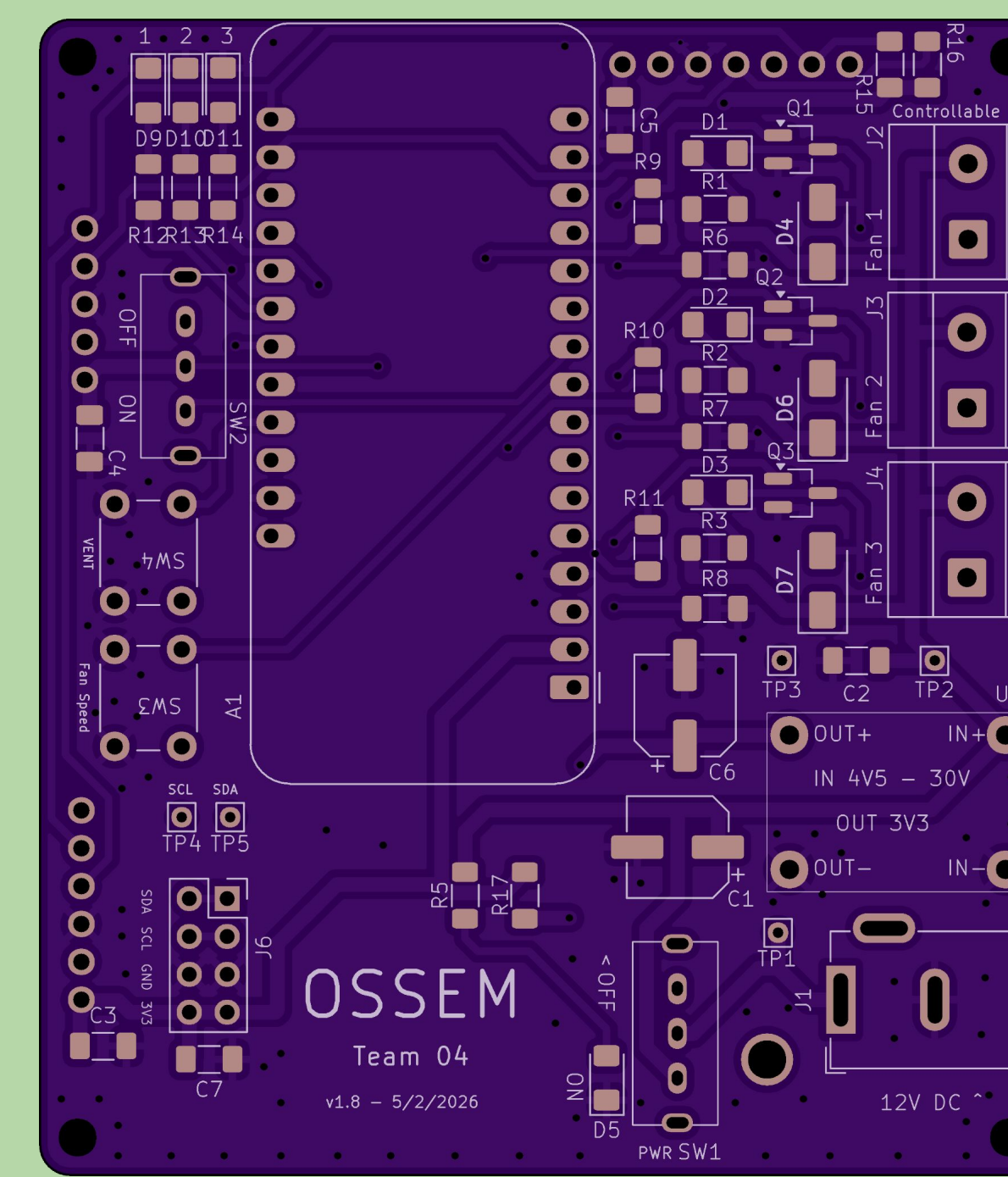


Figure 8: Custom PCB

TESTING & RESULTS

- Warm machine for a few minutes prior to brewing.
- Collect baseline data for 30 seconds.
- Pull a shot of espresso with the enclosure in place while simultaneously logging data using the sensor array.
- Collect post-brew data for 30 seconds.
- Remove enclosure and collect data for an additional 30 seconds.

- Successfully measured temperature, humidity, VOCs, CO₂ concentration, and pressure throughout brewing
- Observed correlation between sensor data and PTR-MS measurements
- Began parallel testing across multiple machines

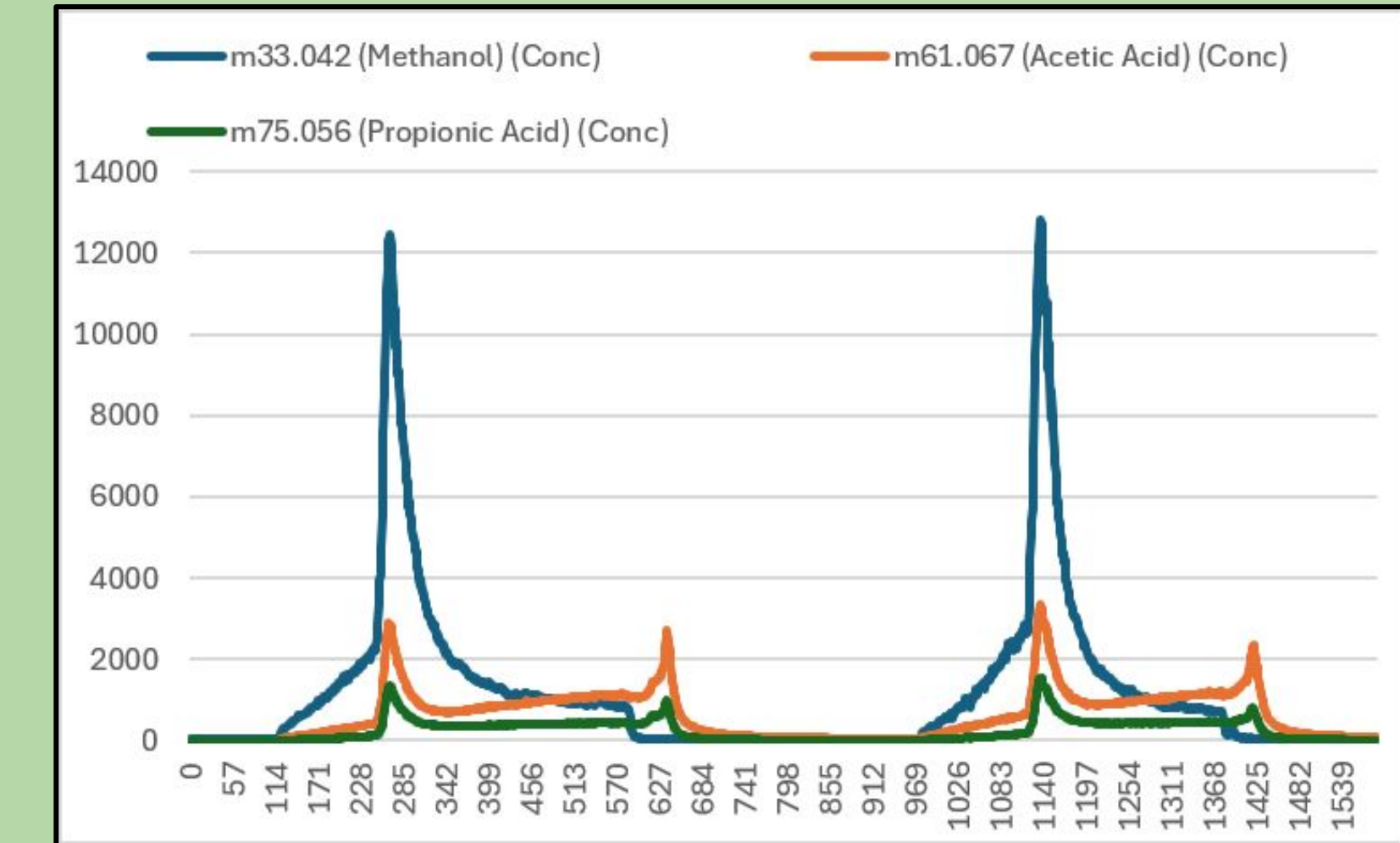


Figure 10: 3D Printed enclosure and PTR-MS

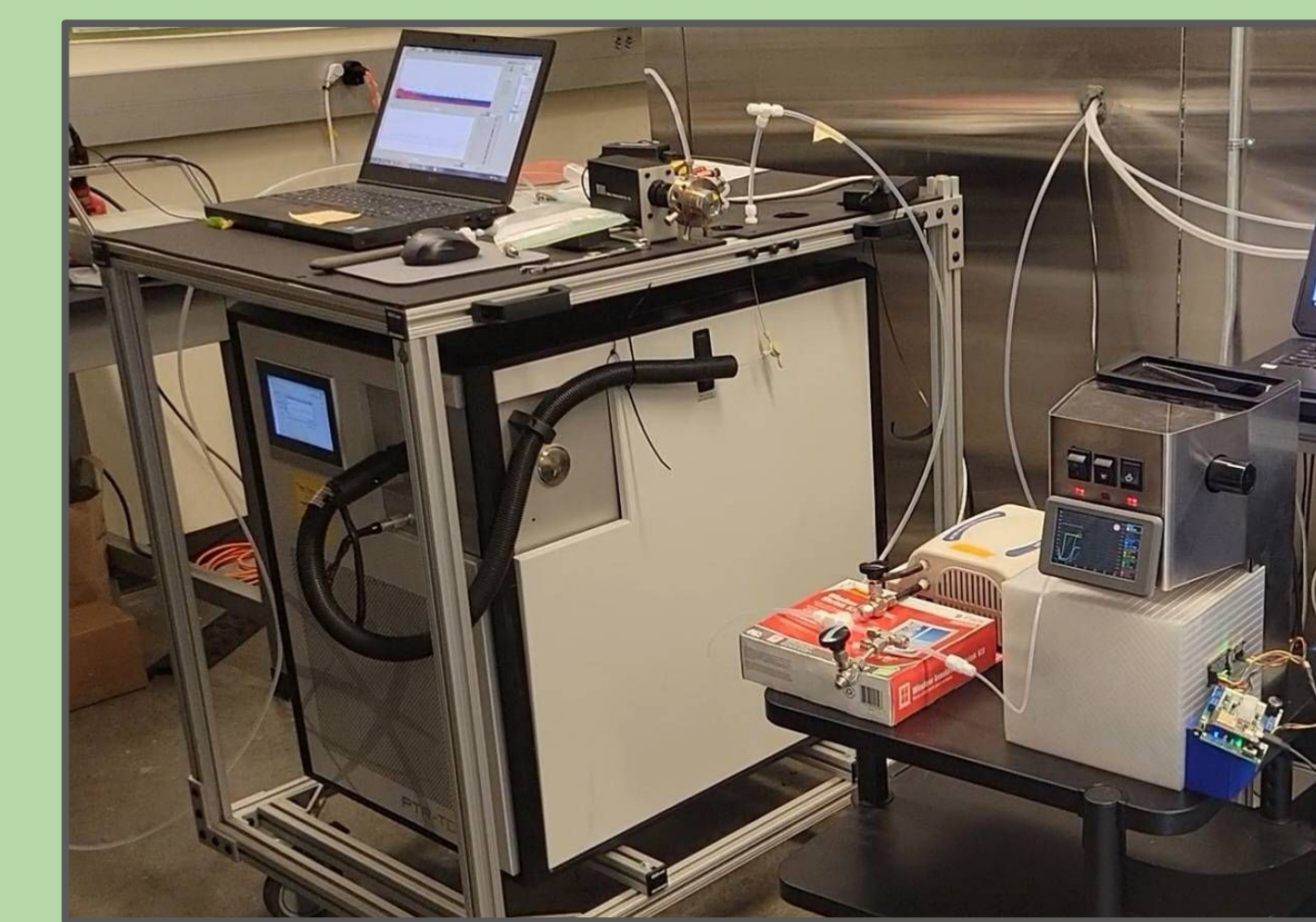


Figure 9: Testing environment with PTR-MS

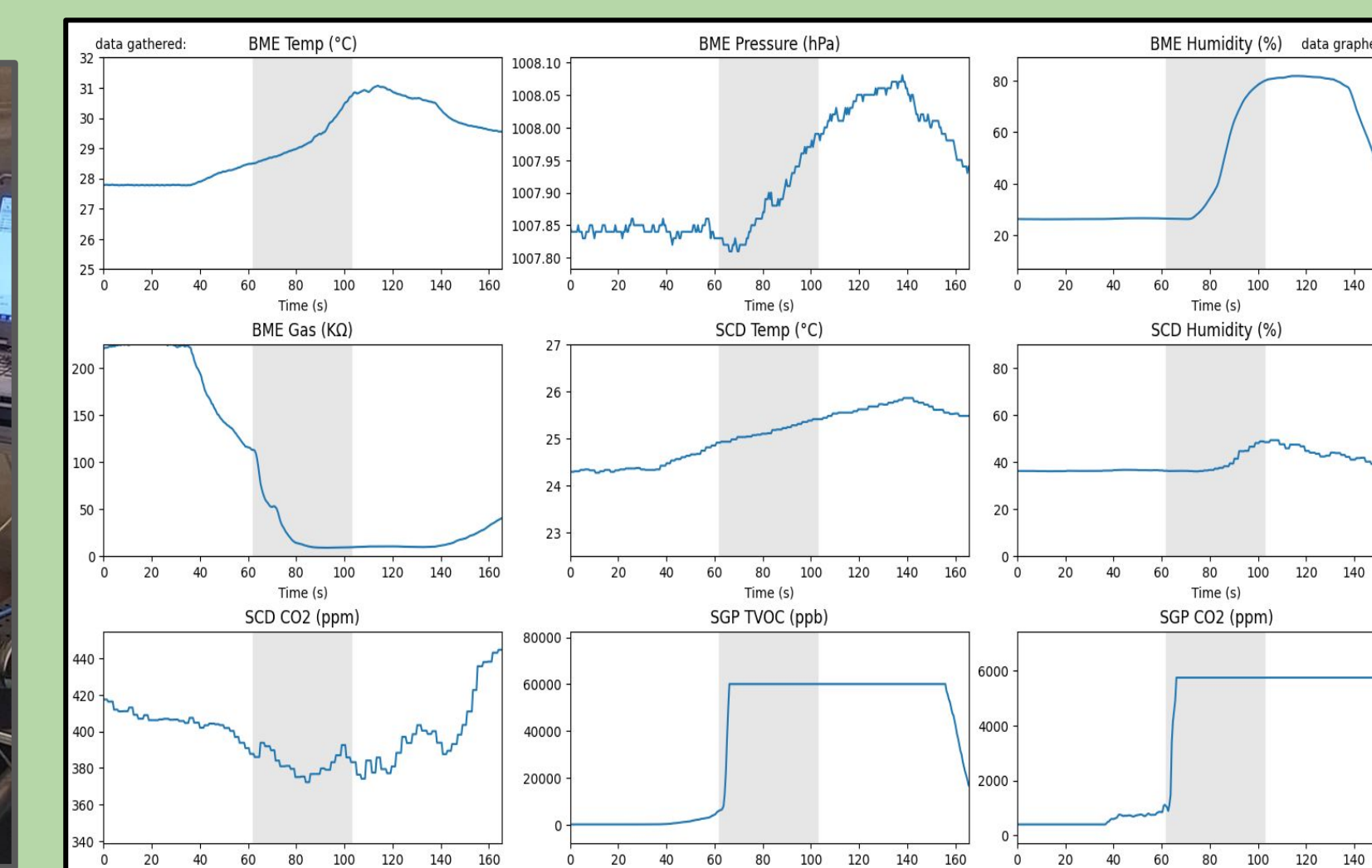


Figure 11: Espresso shot data, fan on slow speed

CONCLUSION

- VOC and CO₂ emissions varied with roast age and brewing conditions.
- Methanol, Acetic Acid, Propionic Acid were among the primary compounds observed.
- Measurement speed and accuracy were limited by sensor design.
- Noisy environments create noisy data!

FUTURE WORK

- Concurrent and post-shot regression analysis of data
 - Integrating noise signals and environmental conditions
- Further system integration, including networking and onboard visualization.